HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026 · PHASE 1

GlucoTwin

A patient digital twin that warns of glucose spikes two hours ahead by fusing health records with wearable data — and tells the doctor what would change the outcome.

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THE PROBLEM

136 million Indians have prediabetes. Most of them are still reversible.

101M
living with diabetes

136M
with prediabetes — still modifiable

2×
times a year a doctor sees them

ICMR-INDIAB national survey, *Lancet Diabetes & Endocrinology*, 2023.

THE GAP

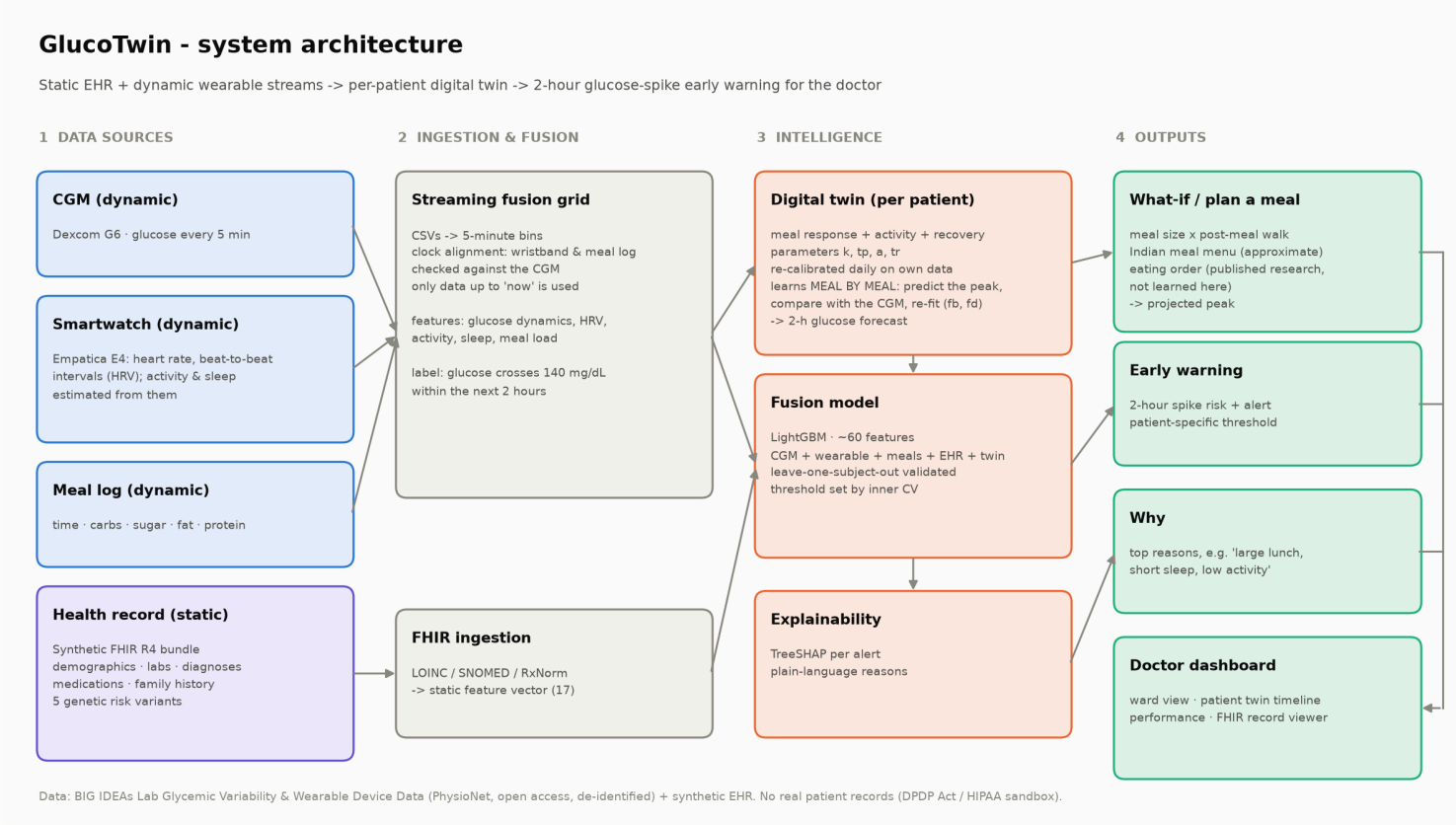
A CGM tells you about a spike. It just tells you too late.

- A doctor sees the patient **twice a year**, and gets one HbA1c — an average over three months.
- A continuous glucose monitor reports the spike **once it is already happening**. By then they have already eaten.
- A simple CGM trend alarm does warn early — at **6.8 false alarms per day**. Nobody keeps listening to that.

Can we warn two hours ahead — and say what to do about it?

THE SYSTEM

Static record + live sensors → a twin per patient



DATA PROVENANCE

What is real, and what is not

STREAM	SOURCE	REAL OR SYNTHETIC
CGM, heart rate, HRV, food logs	BIG IDEAs Lab, PhysioNet (open access)	**Real**, de-identified
Sex, HbA1c	same, Demographics.csv	**Real**, de-identified
Age, labs, diagnoses, medications, family history, genetics	Generated by `src/ehr.py` as FHIR R4	**Synthetic**
Eating-order effect	Shukla et al., *Diabetes Care* 2015	Published research
Indian meal menu	Typical servings	Approximate

16 participants · 8–10 days each · 38,115 five-minute rows · 22% followed by a spike. No real patient record is used anywhere (DPDP Act / HIPAA sandbox).

THE UNGLAMOROUS PART

Three devices. Three clocks.

- **Wristband vs CGM.** Heart rate and glucose are both quiet at night. The shift that lines up the two 24-hour patterns is the offset. Here: **0 hours** — they agreed.
- **Meal log vs CGM.** Glucose should rise 40–80 min after carbs. **Participant 016 was logging meals 105 minutes late** — corrected.
- **Coverage. Participant 015's entire food log falls outside their CGM recording** → marked *unavailable*, meaning "unknown", not "never ate".

Tested on a deliberately broken copy: every offset recovered exactly, AUROC back to within **0.01**.

FEATURE DESIGN

A feature at time T may only use data from time T or earlier

- **~60 features** — CGM dynamics, wearables, sleep, meals, and 17 from the health record.
- **Person-relative wearables.** Resting heart rate varies more *between* people than *within* one. Raw values teach a model *who* someone is — which transfers to a new patient as nothing.
- **Right-closed bins.** A bin labelled 12:05 holds data up to and including 12:05, so a prediction "at 12:05" has never seen 12:06.
- Every rolling statistic is past-only — even the resting heart rate behind the activity estimate.

This is the difference between a glucose model that works and one that is cheating without its author realising.

THE TWIN

Four numbers are the patient

- **k** — carb sensitivity: mg/dL per gram of net carbs
- **tp** — minutes to peak after eating
- **a** — how much activity helps them
- **tr** — how fast they recover

glucose = baseline
+ **k** × carbs, peaking at **tp**
− **a** × recent movement
+ leftover, fading over **tr**

Starts at the population average, then **re-fitted every day on that patient's own past data only**. Only its *forecasts* feed the ML model — the four parameters are constant per person and would act as an ID card.

VALIDATION

Every number comes from a model that never saw that patient

- **Leave-one-subject-out.** Train on 15, test on the 16th, repeat 16 times.
- Random row splits put the same person in train and test and **inflate every score**. Don't.
- **Alert thresholds chosen by inner cross-validation on the training patients only** — nothing about the test patient leaks into the model or the alarm.

RESULTS · LEAVE-ONE-SUBJECT-OUT

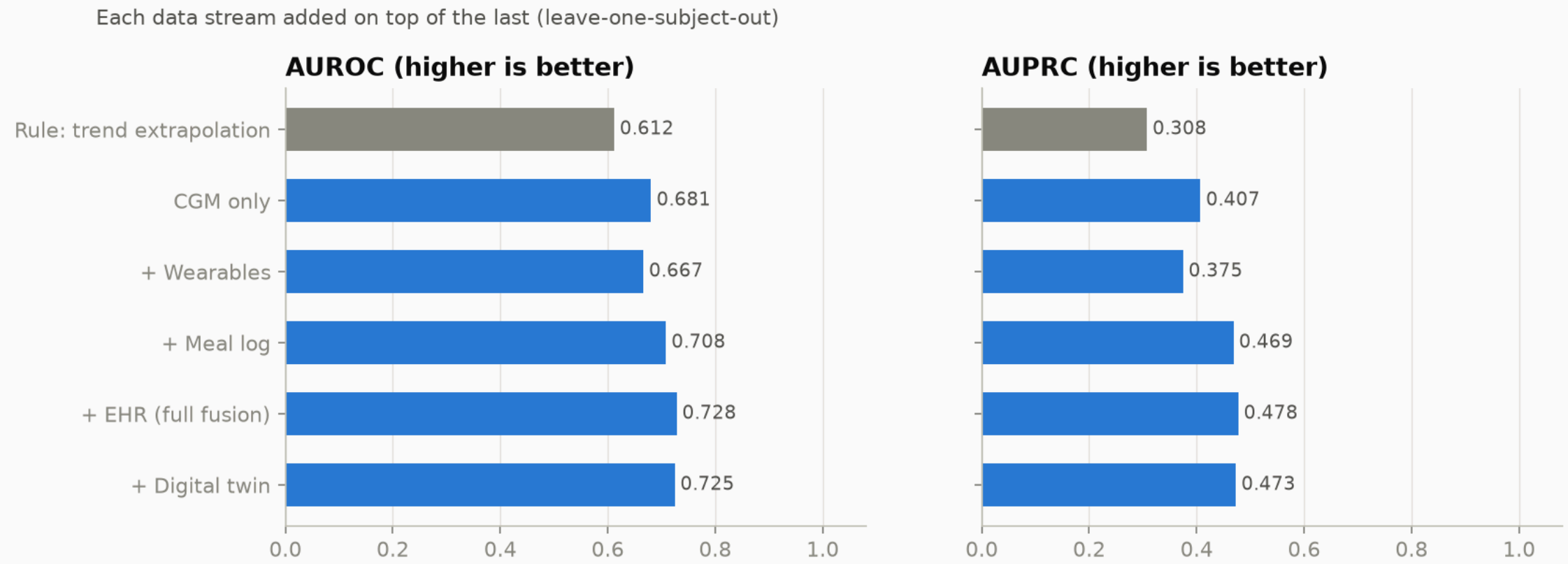
Each row adds one more data stream

DATA USED	AUROC	AUPRC	SPIKES WARNED	MEDIAN LEAD	FALSE ALARMS/DAY
Rule: CGM trend extrapolation	0.612	0.308	99%	75 min	6.8
CGM only	0.681	0.407	91%	10 min	1.7
+ Wearables	0.667	0.375	88%	15 min	2.3
+ Meal log	0.708	0.469	88%	45 min	2.3
+ EHR (full fusion)	**0.728**	**0.478**	**86%**	**45 min**	**2.0**
+ Digital twin forecasts	0.725	0.473	89%	45 min	2.2

AUPRC baseline (random) = **0.22**. Original locked run reported 0.733; re-running on LightGBM 4.7 / pandas 3.0 gives 0.728 — gradient boosting is not bit-reproducible across versions.

RESULTS

Fusion helps. Wearables alone don't — and I'll say why.



Adding wearables drops the score because the accelerometer files (~13 GB) weren't downloaded — activity is estimated from heart rate, which also rises with stress and after meals. They still contribute inside the full fusion.

LIVE DEMO

The ward, at 8pm on day three

- Switch to the dashboard → **Ward overview** → sidebar: **day 3, 20:00**
- **2 alerts · 5 watch · 2 already above 140**
- **P-009, risk 80%** — reason: *"Twin-projected glucose in 1 h — 142 mg/dL"*, 47 g carbs ten minutes ago
- Reasons are **SHAP values** from the model that never saw that patient, in plain language
- Then: cohort filter → **Women 9 / Men 7**

NEW IN THIS BUILD

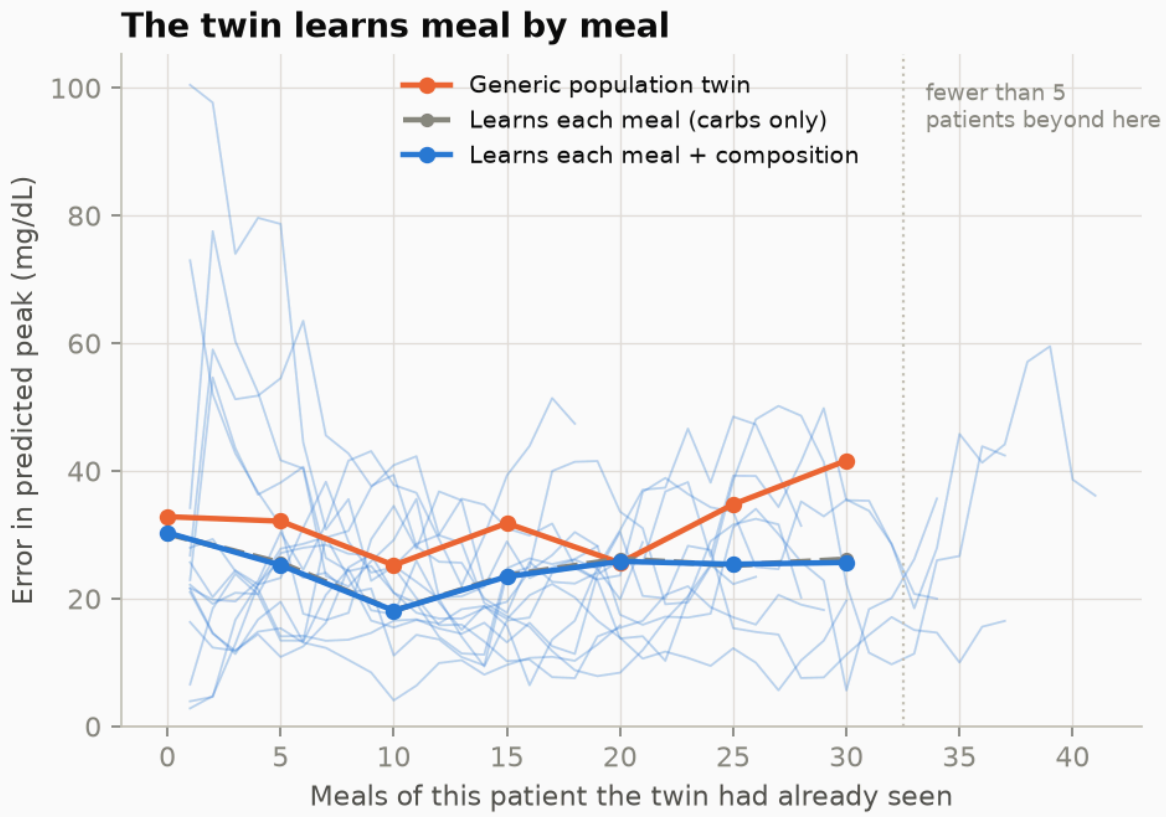
A twin that learns from every meal

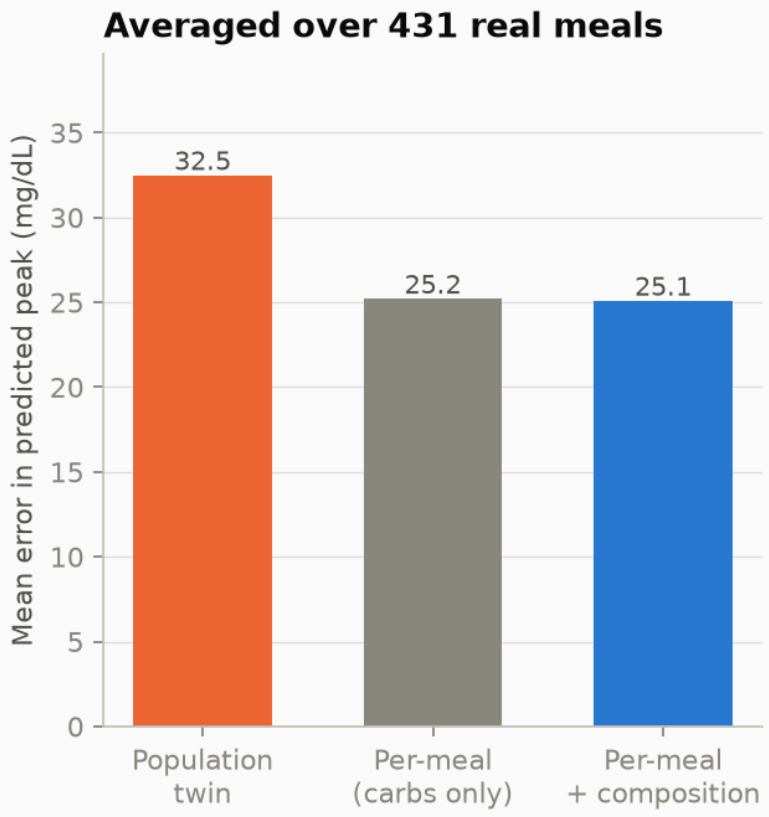
- The daily twin re-fits once a day. But a patient doesn't live in days — **they live in meals**.
- **Before** each meal: predict the peak, from parameters fitted on that person's **earlier meals only**.
- **After** the meal: compare with the real CGM peak.
- **Then** re-fit, so the next meal is predicted by a slightly better twin.

No leakage, and it is **tested, not asserted**: a script deletes future meals and future glucose and checks neither the forecast nor the fitted parameters move. **192 + 53 probes, all unchanged.**

RESULTS · 431 REAL MEALS, 15 PARTICIPANTS

The gap widens the more of you it sees





32.5

mg/dL — generic population twin

25.2

mg/dL — learns each meal (−22%, CI 4.1–10.6, helped 13/15)

+0.12

mg/dL from food composition — CI includes zero. **Did not help.**

LIVE DEMO

"If she eats the dal before the rice, she won't spike"

- Dashboard → **Patient twin** → **P-005** → day **3**, **16:00** → **Plan a meal** → Rice + dal + sabzi
- The alarm says **24%, green, stable**. Nothing here needs a doctor.
- **Rice first → 146.** Crosses 140. **Dal and sabzi first → 135.** Stays under.
- Largest safe portion: ~**115%** of this plate, eaten vegetables-first
- Then: **Build your own** / **Describe it in words**, and **After the meal** → meal 7

NOT LEARNED FROM THIS DATA

Eating order is a published constant, and the UI says so

- The BIG IDEAs food logs record **what** and **when** — never **in what order**. There is nothing here to learn it from.
- Shukla AP et al., *Diabetes Care* 2015;38(7):e98–e99 — crossover, **n = 11** with type 2 diabetes, reporting **29–37%** lower glucose when vegetables and protein came first.
- I apply **25%** — deliberately below their figure, because eleven people eating a Western meal is not this cohort.
- One named constant: `config.ORDER_PEAK_MULTIPLIER`. Change it or switch it off.

WOMEN'S-HEALTH MODULE

Insulin sensitivity falls in the luteal phase — and this cohort can't show it

- The same meal peaks higher in the second half of the cycle. For prediabetes screening in India, where most of those screened are women, that matters.
- **But the BIG IDEAs study enrolled only post-menopausal females.** All 9 women here are post-menopausal. **None has a cycle.**
- So it ships as a **labelled capability demonstration**, behind a warning, defaulting to "Not applicable".
- Demo effect: vegetables-first goes **135 → 141**. The same advice stops working.

The dataset that *would* let the twin learn this: **mcPHASES** (PhysioNet) — 42 women, ~3 months each, daily LH/E3G/PdG assays with labelled phases. In future work.

A NEGATIVE RESULT, REPORTED

I re-ran everything on the 9 women alone

DATA USED	AUROC — ALL 16	AUROC — 9 WOMEN
CGM only	0.681	0.607
+ Meal log	**0.708**	**0.667 ← best here**
+ EHR (full fusion)	0.728	0.654
+ Digital twin	0.725	0.655

Full fusion falls to **0.654**, and the health record makes it *worse*. That's a **cohort-size effect, not a finding about women** — nine people, one with no usable food log. It's why the headline keeps all 16.

LIMITATIONS

Arguing against my own project

- **Sixteen people.** Feasibility, not clinical validity.
- **The health record is synthetic** apart from sex and HbA1c — so the EHR result demonstrates the *pipeline*, not real records.
- **Activity and sleep are estimated** from heart rate, not accelerometry or polysomnography.
- **Unlogged meals can't be anticipated.**
- **Eating order and cycle phase are published constants**, labelled on screen every time.
- **Food composition didn't help** — reported as the negative result it is.

WHAT IT DOES SHOW

Not "she will spike" — but "if she eats the dal first, she won't"

0.612→0.728
AUROC, glucose alone → full fusion

6.8→2.0
false alarms per patient-day

−22%
peak error, twin learning meal by meal

Future work: real EHRs through ABDM/FHIR consent flows · a cycle-aware twin learned from mcPHASES · federated on-device training · hypoglycaemia alerts.

Decision support, not a prescription. Not a medical device. Deployment would need prospective validation and regulatory review.

THANK YOU

GlucoTwin

Code, data provenance, every result and the leakage test are in the repository.

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github.com/varshabachar/Uno_RVUniversity

Data: BIG IDEAs Lab Glycemic Variability & Wearable Device Data, PhysioNet (ODC-By)